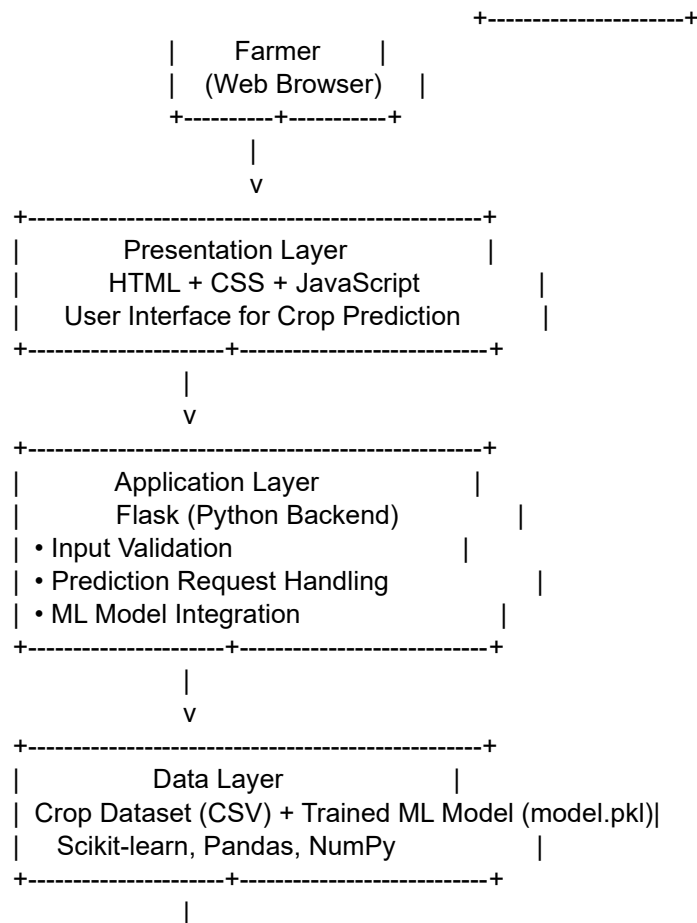


Solution Architecture

Date	2 July 2026
Team ID	xxxxxxx
Project Name	OptiCrop – Smart Agricultural Production Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	Provides the user interface for entering soil nutrients and weather details and displays the recommended crop.	HTML, CSS, JavaScript
<i>[API Gateway]</i>	Receives user requests, validates the input, and forwards them to the Machine Learning model for prediction.	Flask (Python)
<i>[Microservices]</i>	Executes the crop prediction logic by loading the trained model and processing user input.	Scikit-learn, Joblib, Pandas, NumPy
<i>[Database]</i>	Stores the crop dataset used for training and the trained Machine Learning model used for prediction	CSV Dataset, model.pkl